Venkata Lakshmi Parimala Pasupuleti

Business Analytics · Data Analysis · Financial Data Management · Process Improvement · Data Governance · Regulation Compliance · Tableau · SQL

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PROFESSIONAL SUMMARY

Business analyst and IT professional with 8+ years translating complex business requirements into **data-driven solutions** - delivering analytical dashboards, financial data pipelines, and process improvement initiatives across regulated, compliance-heavy environments. At Entrata (PropTech SaaS, 100K+ MAU), led **financial planning and management** for affordable housing operations: HUD payments, rental subsidies, budget-critical HAP allocations, and AMI-based pricing - ensuring **regulation compliance** with zero violations. Built **advanced analytical reports and dashboards** (Tableau, ggplot2) supporting executive decision-making; optimized 200+ SQL queries for **data validation** and reporting accuracy across all client environments. Skilled in **data governance, enterprise data management standards, stakeholder collaboration, and project management** across 120+ Agile sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Business Analytics & Decision Analytics | Data Analysis & Reporting | Financial Planning & Management | Financial Modeling & Cost-Benefit Analysis | Process Improvement & Performance Improvement | Data Governance & Data Management Standards | Regulation Compliance | Dashboards & Analytical Reports (Tableau) | Data Validation & Data Quality | SQL & Databases | Python & Spreadsheet Analysis | Project Planning & Project Management | Stakeholder Collaboration & Business Case Development | Information Technology Coordination

EXPERIENCE

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Research Associate · Business Analytics · Data Visualization · Stakeholder Reporting

- Delivered **advanced data analytics and decision support** to faculty stakeholders: built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets achieving $R^2 = 0.977$ across 190+ countries — rigorous **statistical analysis**, hypothesis testing, and **data validation** pipelines driving research strategy recommendations.
- Designed **Tableau and ggplot2 leadership dashboards** and **analytical reports** tracking KPI metrics across two research cohorts — improved stakeholder comprehension by 40%; performed exploratory analysis and **data quality** monitoring across 3 active research pipelines.
- Applied Isolation Forest anomaly detection to automate **data validation** and integrity checks — surfaced data quality issues that manual review missed; improved downstream analytical reliability.
- Led **project planning and execution** for data infrastructure initiatives; coordinated deliverables, facilitated stakeholder meetings, and mentored 600+ students in SQL, Python, data analysis, and database design.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Financial Data Management · Regulation Compliance · Process Improvement · IT Coordination

- **Led financial planning and management** for affordable housing operations: HAP allocation, rent pricing strategy (AMI-based pricing tiers, gross rent updates), HUD payments, rental subsidies, tax credit calculations, and ledger management — analytical, quantitative **business analytics** ensuring compliance with federal and state regulations across 100K+ unit portfolios; collaborated cross-functionally with compliance, operations, and product stakeholders on **business case development** and policy recommendations.
- Developed **detailed analytical reports, dashboards, and design specifications** for financial and compliance operations: analyzed AMI threshold distributions across 1.6M HUD/Tax Credit income records (5 years, 5 US states); built occupancy risk dashboards tracking recertification compliance rates, budget exposure, and performance metrics — supporting executive **decision analytics** and growth opportunity assessments.
- **Optimized 200+ SQL queries** (30% performance gain) through execution plan tuning, indexing, and database partitioning; built **data validation** and audit tools — log tables, traceability pipelines, and data monitoring dashboards ensuring **data management standards** and SLA compliance across all client environments.
- Drove **process improvement and performance improvement**: eliminated 80% of manual operational workload via Python automation pipelines; led **microservices architecture** redesign (50% scalability improvement, 30% deployment time reduction); converted 20% of product modules into service-based architecture — initiated with formal **cost-benefit analysis** and business case presented to leadership.
- Ensured **regulation compliance and data governance**: achieved 100% PHI compliance on AWS S3 migration with Libsodium-encrypted pipelines; enforced **enterprise data management standards** across all data insertion and retrieval operations; zero security incidents; zero compliance violations across 6+ years. Mentored 5+ junior analysts; conducted 300+ peer code reviews.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Business Analytics · Compliance Automation · Data Governance

- Built a **business analytics and compliance pipeline** that profiles operational data against regulatory standards, applies **data validation** logic, and routes decisions with grounded, cited justifications — enforcing **data governance** and management standards across multi-step analytical workflows aligned with organizational compliance requirements.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Financial Data Pipelines · SQL · Regulation Compliance

- Built financial data pipelines for affordable housing compliance (PHP, PostgreSQL); designed SQL import scripts processing 1.6M HUD/Tax Credit income records; enforced **regulation compliance** and **data management standards** across document storage and database architecture; created 56 state-specific compliance certification documents with merge fields.

PORTFOLIO PROJECTS

AI on FHIR — Healthcare Data Analytics Platform with HIPAA Compliance & Data Governance

Healthcare Data · Regulation Compliance · Data Governance

2024 · Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings · github.com/venkatalakshmiparimala/AI_on_FHIR

- Built an NLP-driven **healthcare data analytics platform** on FHIR (Fast Healthcare Interoperability Resources) standards — translating natural language queries to structured medical record retrieval; reduced manual clinical data retrieval time ~60% while enforcing **HIPAA-compliant RBAC, data governance, and data privacy** controls across structured health records.
- Implemented **enterprise data management standards** including role-based access, metadata standards, and regulatory compliance controls — designed to operate within health system data governance frameworks; collaborated with clinical and technical stakeholders to define data access objectives and reporting requirements.

Leasel — Analytical Reporting & Data Governance Pipeline for Regulatory Compliance

Data Governance · Analytical Reports · Regulation Compliance

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed a **data governance and analytical reporting pipeline** enforcing **regulation compliance** on 1,240 residential compliance documents (5 US states + federal) — 93.6% precision across 10 compliance rules; automated **data validation, data profiling**, and missing-clause detection; 100% citation correctness; zero unsupported determinations.
- Facilitated **data governance activities**: outlined objectives, standardized metrics and definitions, enforced **enterprise data management standards**, and designed human-review gates ensuring every regulated financial or eligibility determination is auditable and traceable to source documentation.

CrashClarity — Large-Scale Data Analysis & Business Intelligence Dashboard (104K Records)

Business Analytics · Decision Analytics · Dashboards

2025 · Python, R, scikit-learn, Random Forest, Logistic Regression, Tableau, GitHub Pages · View Project

- Performed end-to-end **business analytics and decision analytics** on 104K+ records — data profiling, cleansing, feature engineering, and **financial modeling**-style impact assessment across behavioral and environmental variables; achieved 85%+ model accuracy; deployed interactive **business intelligence dashboard** for stakeholder reporting and competitive assessment of risk factors.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

Coursework: Statistical Analysis, Data Mining, Financial Modeling, Data Visualization, Database Systems, Data Governance, Applied Business Analytics

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Business & Decision Analytics: Business Analytics, Decision Analytics, Business Intelligence, Data Analysis, Financial Planning, Financial Management, Financial Modeling, Cost-Benefit Analysis, Business Case Development, Process Improvement, Performance Improvement, Procurement Support, Budget Analysis

Data & Reporting: SQL (PostgreSQL, MySQL, Oracle), Python (Pandas, NumPy, scikit-learn), R, Spreadsheets (Excel), Databases, Report Writing, Analytical Reports, Dashboards, Tableau, Power BI, Matplotlib, Seaborn, ggplot2, ETL, Data Pipelines

Governance & Compliance: Data Governance, Enterprise Data Architecture, Data Management Standards, Metadata Standards, Regulation Compliance, Regulatory Compliance, Data Validation, Data Quality Management, Data Profiling, HIPAA Compliance, PHI Data Handling, Audit Trails

Project & Stakeholder Management: Project Planning, Project Management, Agile/Scrum, Stakeholder Collaboration, Cross-Functional Collaboration, Information Technology Coordination, Supervisory Experience (5+ engineers mentored), 300+ Code Reviews

Engineering & Infrastructure: Python, FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, CI/CD, PostgreSQL, MySQL, Microservices, Apache Spark, PySpark, LangChain, LangGraph, RAG Pipelines